

Setup system

Setup the system and get files from the internet.

In the following code, we will download the necessary files from the internet.

Make videos I: Pick Find character images

Assignment: Find character images

1. Click on your Folder icon on the Left Hand Side.
2. Click on drive.
3. Click on MyDrive.
4. Click on Race-2025.
5. Click on GraphSpeeds2-main.
6. Look at the different images.
7. Write down the names of the image files that you want to use.

Make videos II: Create tables for your characters

Assignment: Create the tables and equations for your characters

1. Enter time and distance units in column_labels.
2. Enter time_values and distance_values for each value.
3. Enter names and equations for each character.

```
In [2]: # Prepare the labels for everyone
column_labels = ["x (seconds)", "y (feet)"] # <- FIX units

# Table #1 values:
time_values    = [0, 1, 2, 5.5, 11] # <- ENTER x values
distance_values = [0, 4.5, 9, 25, 50] # <- ENTER y values

# Create your table:
char1 = table(x=time_values, y=distance_values, column_labels=column_labels)
char1.plotTable()

# Explicitly run choreo_get_chrome here as the error message still suggests it,
# even after setup cell's kaleido_get_chrome and un/reinstall.
!choreo_get_chrome
char1.saveImage("char1.png")

# Table #2 values:
time_values    = [0, 1, 2, 4.5, 9] # <- ENTER x values
distance_values = [0, 5.5, 11, 25, 50] # <- ENTER y values

# Create your table:
```

```

char2 = table(x=time_values, y=distance_values, column_labels=column_labels)
char2.plotTable()
char2.saveImage("char2.png")

# Table #3 values:
time_values      = [0, 1, 2, 3, 6] # <- ENTER x values
distance_values  = [0, 8.3, 16.6, 25, 50] # <- ENTER y values

# Create your table:
char3 = table(x=time_values, y=distance_values, column_labels=column_labels)
char3.plotTable()
char3.saveImage("char3.png")

# Table #4 values:
time_values      = [0, 1, 2, 2.5, 5] # <- ENTER x values
distance_values  = [0, 10, 20, 25, 50]
# Create your table:a
char4 = table(x=time_values, y=distance_values, column_labels=column_labels)
char4.plotTable()
char4.saveImage("char4.png")

# Merge all of the tables
tables_list = [char1, char2, char3, char4]

# Line equations of tables
legend_labels = [
    'two_thumbs: y = 1*x', # <- Modify name and the equation here.
    'hello kitty: y = ?*x', # <- Modify name and the equation here.
    'chex: y = ?*x', # <- Modify name and the equation here.
    'kingpin: y = ?*x'] # <- Modify name and the equation here.

# Plot tables
plotTablesLines(tables_list,
    fig_title = 'The Race', # title on top
    x_label = 'Elapsed time (hours)', # appears along x-axis
    y_label = 'Distance in miles', # appears along y-axis
    legend_title = 'Journey', # title on the right hand side
    legend_labels = legend_labels, # text describing the table
    img_name = 'race.png') # Saves under race.png.

```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	4.50000
2.00000	9.00000
5.50000	25.00000
11.00000	50.00000

/bin/bash: line 1: choreo_get_chrome: command not found
Wrote char1.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	5.50000
2.00000	11.00000
4.50000	25.00000
9.00000	50.00000

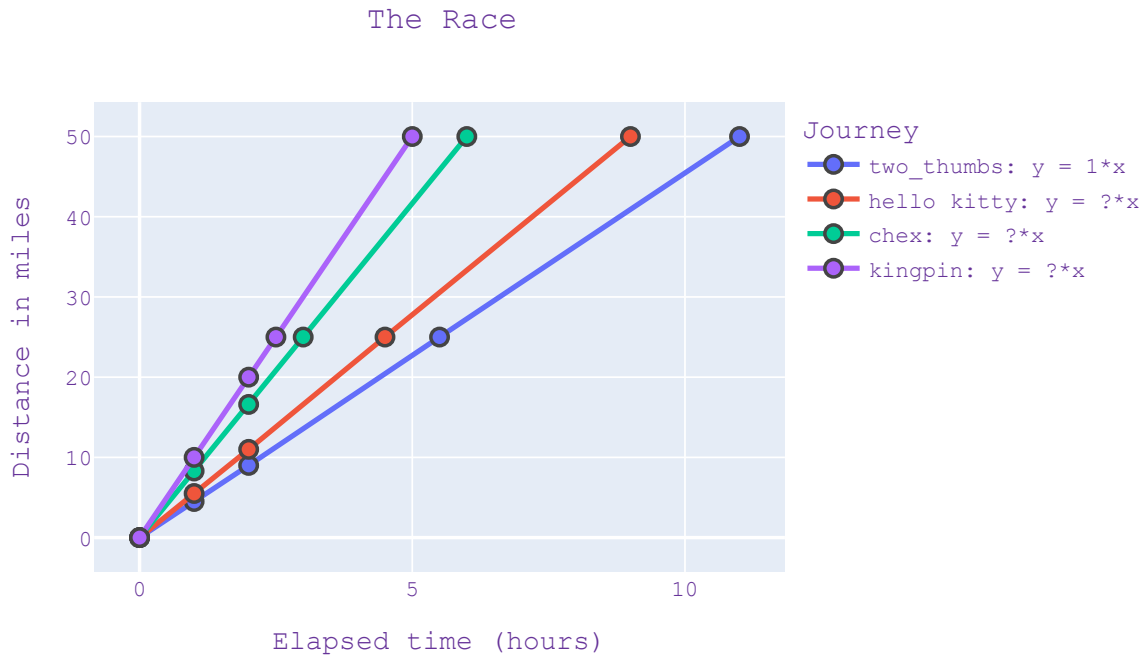
Wrote char2.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	8.30000
2.00000	16.60000
3.00000	25.00000
6.00000	50.00000

Wrote char3.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	10.00000
2.00000	20.00000
2.50000	25.00000
5.00000	50.00000

Wrote char4.png



Make videos III: Specify names, images, and speeds

Assignment: place your characters

1. ENTER the names of the characters for your video.
2. ENTER filenames for each character in your video.
3. ENTER the speeds for your characters.

```
In [3]: # Enter the names of your characters:
names = ["two thumbs", "chex", "kingpin", "hello kitty"]

# Enter the file names of your images:
images = ["race-car.jpg", "block-man.jpeg", "spiderman.png", "hello-kitty.jpeg"]

# Enter the speeds:
speeds = [4.54, 5.55, 8.33, 10 ]

# Directory:
directory = "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/"
images_list = [directory + filename for filename in images]

# Save all of the data:
data = [names, images_list, speeds]
```

Make videos IV: Make and experiment with your

video

Your video will be stored to the video file `race.mp4` or a filename that you choose.

To make a video, one time unit (e.g., one hour) corresponds to one second of video.

If you set the `simulation_speed=2.0`, then one second will correspond to two time units (e.g., two hours).

For each second, the video contains a number of images. This is controlled by `fps` which means number of frames per second.

Assignment: experiment with your video

1. Enter the race distance, duration of the race, and video title.
2. Enter the units.
3. Change `simulation_speed` to 2.0. What do you observe?
4. Change `fps` to 20. What do you observe?
5. Save the video to a different file name.

```
In [4]: # Experiment with the video
sim_video = simulationVideo(data = data,          # characters
                             duration = 16,       # ENTER duration
                             race_distance = 50,  # ENTER race distance
                             vid_title = "Race")  # video title

sim_video.set_video(video_name="race.mp4",        # new filename
                    fps=10,                       # frames per second
                    simulation_speed=1.0)         # speed of the simulation

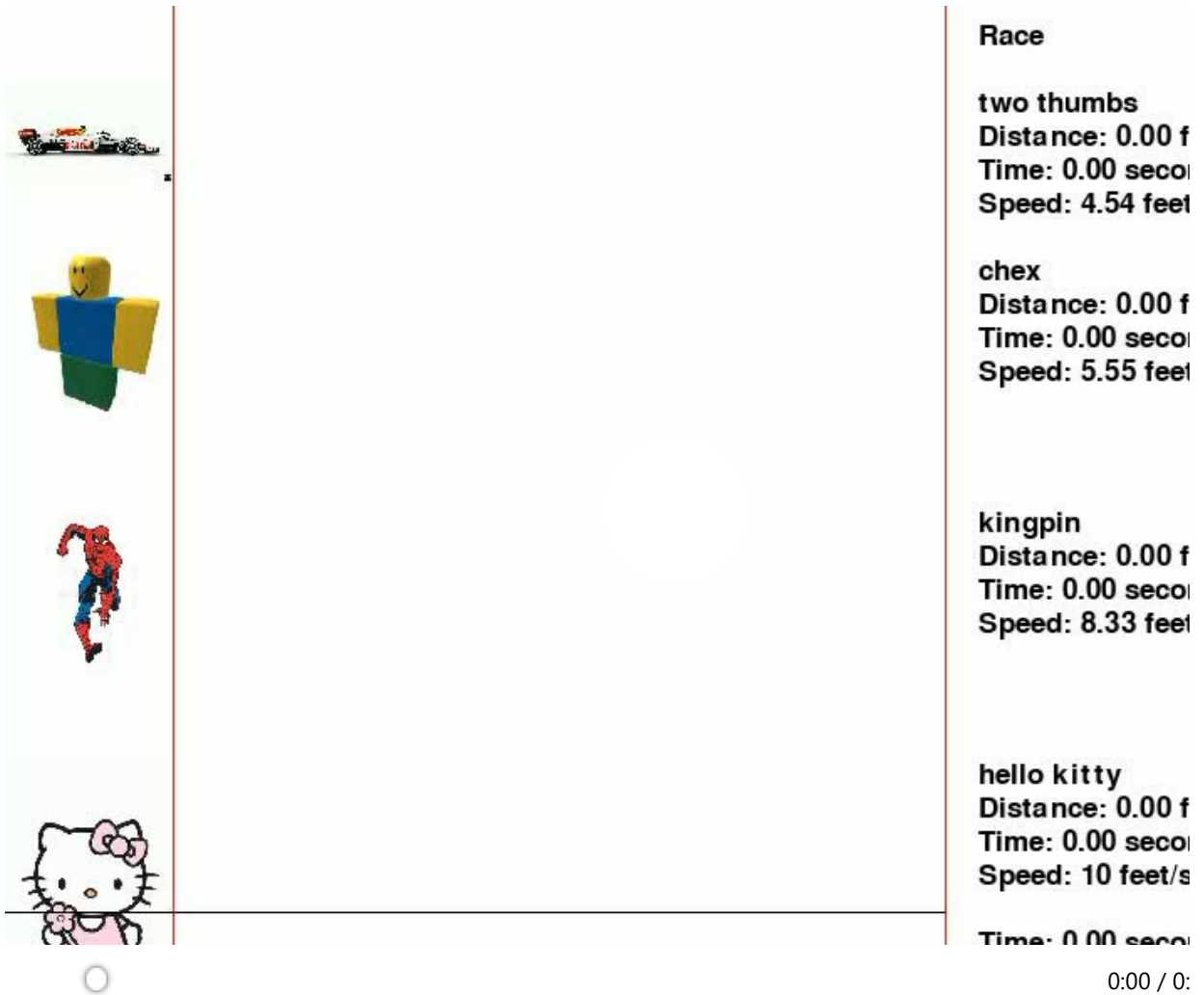
sim_video.set_units(distance_string="feet",        # ENTER distance unit.
                    time_string="seconds",         # ENTER time unit.
                    speed_string="feet/second")    # ENTER speed units.

# Create and save your video:
video = sim_video.create_video()
video.ipynon_display()
```

```
File 'race.mp4' has been removed.
Videwriter initialized successfully.
video file = race.mp4 closed.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4
```

```
Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

Out[4]:



Make videos V: Make title and explanatory text images

You can create many text frames and save them into different frames.

This is done using `saveImage("filename.png")` as given below:

```
text = ""
First line.
Second line.
Third line.
textImage(frame1.png", text)
```

Assignment: Create text images

Fill out the explanatory text images.

```
In [5]: # Create Title and group name here
text = ""
Session 2 Group A
the hellcats
""
textImage("frame1.png", text)
```

```

# Who is racing?
text = """ Hello Kitty, Spider man, an F1 car and Noob are racing
"""
textImage("frame2.png", text)

# Why they are racing?
text = """ They are racing because they wanted to see who the fastest
...
"""
textImage("frame3.png", text)

# Character 1 equation and explanation:
text = """ Character 1 equation:
 $y = 4.54 * x$ 
...
"""
textImage("frame4.png", text)

# Character 2 equation and explanation:
text = """ Character 2 equation:
 $y = 5.55 * x$ 
...
"""
textImage("frame5.png", text)

# Character 3 equation and explanation:
text = """ Character 3 equation:
 $y = 8.33 * x$ 
...
"""
textImage("frame6.png", text)

# Character 4 equation and explanation:
text = """ Character 4 equation:
 $y = 10 * x$ 
...
"""
textImage("frame7.png", text)

# Explain the winner:
text = """ The CAR lost... sad
...
"""
textImage("frame8.png", text)

# Victory page:
text = """ hello kitty won because she was the fastest
...
"""
textImage("frame9.png", text)

```

Make videos VII: Make the final video

You can make the final video by combining images and videos together.

Assignment: Make final video

Verify that your video satisfies the requirements of your assignment!

```
In [6]: # Confirm that you have all the image here:
img_list = ["frame1.png", "frame2.png", "frame3.png",
            "race.mp4",
            "char1.png", "frame4.png",
            "char2.png", "frame5.png",
            "char3.png", "frame6.png",
            "char4.png", "frame7.png", "race.png",
            "frame8.png", "frame9.png"]

# Create the video:
video=CreateVideo(video_name="all.mp4",      # This is file name of the final video.
                  file_list=img_list,       # This the list of images and files to combine.
                  fps=10,                  # The number of frames per second.
                  duration=2.5)            # The duration for each image.

# Display the video:
video.ipython_display()
```

```
Opening frame1.png
Opening frame2.png
Opening frame3.png
Opening race.mp4
Opening char1.png
Opening frame4.png
Opening char2.png
Opening frame5.png
Opening char3.png
Opening frame6.png
Opening char4.png
Opening frame7.png
Opening race.png
Opening frame8.png
Opening frame9.png
video: height = 600 width = 800
File 'all.mp4' has been removed.
VideoWriter initialized successfully.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4

Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

Out[6]:

Session 2 Group A
the hellcats



0:00 / 0:

SAVE YOUR VIDEOS AND GOOGLE COLAB NOTEBOOKS

```
In [7]: !jupyter nbconvert --to html /content/drive/MyDrive/Colab\ Notebooks/GraphSpeeds_C2_G1.ipynb
```

```
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab Notebooks/GraphSpeeds_C2_G1.ipynb to html
```

```
[NbConvertApp] Writing 364430 bytes to /content/drive/MyDrive/Colab Notebooks/GraphSpeeds_C2_G1.html
```

```
In [7]:
```